

Robin Duvoisin

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🏠 robinduvoisin.github.io

🌐 [Robin-Duvoisin](#)

EDUCATION

2023-Now	ETH Zürich & University of Zürich, M.Sc. Neural Systems and Computation	5.8/6
2021-2022	ETHZ Exchange Student, B.Sc. Health Sciences and Technology	
2019-2021	EPF Lausanne, B.Sc. Life Sciences Engineering	4.8/6

RESEARCH EXPERIENCE

- 2026 **Master Thesis: Energy-Efficient Neural Codes**, Bern University, Theoretical Neuroscience
Supervisors: Prof. Jean-Pascal Pfister, Dr. Pau V. Aceituno, 8 months
- Developed an information-theoretic framework for receptive field neural coding.
 - Designed a source encoding framework aligned with a previously developed channel encoding model in collaboration with Pau V. Aceituno.
 - Reframed the theoretical problem toward applications relevant to the neuromorphic community.
- 2025 **Visiting Student Researcher**, Stanford University, Department of Bioengineering
Supervisors: Prof. Todd Coleman, Prof. Zhenan Bao, Tomasz Zaluska, 5 months
- Designed and prototyped a minimally invasive neural electrode for deep-brain recording.
 - Developed bistable kirigami tiling structures in flexible materials enabling post-insertion expansion.
 - Extended kirigami theory to include new designs with a positive Poisson's ratio.
- 2023 **Biomechanics Engineer Intern**, *Autonomyo* (Startup) in collaboration with EPFL Rehasist Group
Supervisors: Dr. Amalric Ortlieb, Dr. Mohamed Bouri, 3 months
- Developed algorithms to extract gait kinematics from motion capture and force sensors.
 - Quantified effects of motor control parameters and neurological conditions on gait symmetry and range of motion.
- 2022 **Research Assistant Intern**, ETHZ, Institute of Neuroinformatics, Birdsong Group
Supervisors: Prof. Richard Hahnloser, Corinna Lorenz, 3 months
- Analyzed local field potentials (LFPs) and spiking activity in songbirds to identify synchronization between Area X and LMAN during sleep.
 - Characterized phase-locked LFP features as biomarkers of motor learning in pitch adaptation experiments.

PUBLICATION & PRESENTATION

- Lorenz, C., Das, A., Centeno, E. G. Z., Yeganegi, H., **Duvoisin, R.**, Ursu, R., Retailleau, A., Giret, N., Leblois, A., Hahnloser, R. H. R., & Ondracek, J. M. (2025). *Sharp Waves, Bursts, and Coherence: Activity in a Songbird Vocal Circuit Is Influenced by Behavioral State*. The Journal of Neuroscience.
- **Duvoisin, R.**, Aceituno, P.V. (2025). *Energy-Efficient Neural Coding under Power-Law Distribution: An Information-Theoretic Perspective*. Poster, Conference on Complex Systems 2025, CoBrain Satellite Workshop.

SELECTED PROJECTS

2024 **Energy-Efficient Neural Coding under Power-Law Distribution,**

Supervisor: Dr. Pau V. Aceituno

- Connected existing energy-efficient neural coding schemes to formal information theory frameworks.
- Developed a probabilistically modulated channel-coding framework to better capture biologically plausible inputs.

2023 **Interconnected Multilayer Model to Highlight Particularities of the Retinal Connectome,**

Supervisor: Prof. Alexandre Bovet

- Modelled the retinal connectome as a multilayer network to isolate cell-type-specific connectivity patterns.
- Applied complex network methods to interpret functional organization in a neural circuit.

2022 **Bachelor Project: Uncovering Sleep Modulation Effects through EEG Signals, ETHZ**

Supervisors: Dr. Caroline Lustenberger, Dr. Manuel C. Dominguez

- Designed an experimental protocol to test how pre-sleep arousal influences memory consolidation.
- Processed EEG data from a finger-tapping task and extracted neural markers related to fatigue.

SKILLS

- Programming Languages: Python, MATLAB, C++
- Programming Libraries: PyTorch, NetworkX, Seaborn, Chronux, EEGLab, Signal Processing Toolbox
- Engineering: AutoCAD (laser cutting, 3D printing), COMSOL (hyperelastic modelling)
- Languages: French (native), English (advanced), German (upper-intermediate)

TEACHING EXPERIENCE

2024 **Teaching Assistant, ETHZ, Computer Vision Lab, 4 months, ~100 students**

- Supervised coding exercises and exam grading for image analysis techniques.

2022 **Mentor for first-year students, EPFL, ~10 students**

- Guided students through coursework during the pandemic, organized group sessions and individual support.

AWARDS & SCHOLARSHIPS

2025 Science Alumni UZH Scholarship, 2500 CHF

2025 UZH Mobility Exchange Grant, 2500 CHF

2021 EPFL Mobility Scholarship, 1500 CHF

2019 Auguste Piccard Award, Best Student award in Chemistry and in Biology

EXTRA-CURRICULAR ACTIVITIES

NSC Student Body	Organised peer-led neuroscience discussions and coordinated ski camps
Film Production	Director of Photography, Screenwriter
Ski Touring	Patrouille des Glaciers (Swiss Alpine race)

REFERENCES	Prof. Jean-Pascal Pfister	University of Bern	jeanpascal.pfister@unibe.ch
	Prof. Todd Coleman	Stanford University	toddcol@stanford.edu